PADHAKU
Mathematics · Class 10 Mathematics



CHAPTER 10

Circles

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

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❓ In-text questions · 10.1–10.3

? **Is there any position of a line with respect to a circle other than non-intersecting line, secant and tangent?**

✔ ANSWER

No. A line and a circle in a plane can have 0, 1 or 2 common points only.

- 0 points → non-intersecting line
- 2 points → secant
- 1 point → tangent

So there is no other possible position.

? **Look at a moving bicycle or cart wheel. Do you see any tangent anywhere?**

✔ ANSWER

Yes. The ground on which the wheel moves is a tangent to the circle representing the wheel; the point where the wheel touches the ground is the point of contact.

In every position, the radius through the point of contact is at right angles to the ground (tangent).

? **In the proof of Theorem 10.1, why must the point Q (on the tangent XY, other than P) lie outside the circle?**

✔ ANSWER

If Q were inside the circle, the line XY would cut the circle at two points and would be a secant, not a tangent.

Since XY touches the circle only at P, every other point of XY lies outside the circle. Hence **OQ > OP** (radius).

? In Fig. 10.6 (iii), PT_1 and PT_2 are the lengths of the tangents from P to the circle. Measure PT_1 and PT_2. Are these equal?

✔ ANSWER

Yes, **$PT_1 = PT_2$**.

The lengths of the two tangents drawn from an external point to a circle are always equal (Theorem 10.2).

? In Example 3, why is ∠TPR + ∠RPO = 90° = ∠TPR + ∠PTR?

✔ ANSWER

- ∠TPR + ∠RPO = ∠TPO = 90°, because TP is a tangent and OP is the radius through the point of contact (Theorem 10.1).
- In right ΔTRP (OT ⊥ PQ, so ∠TRP = 90°), the other two angles add to 90°: ∠TPR + ∠PTR = 90°.

Hence **∠RPO = ∠PTR**.

Exercise 10.1

1 **How many tangents can a circle have?**

✔ ANSWER

A tangent can be drawn at every point of the circle, and a circle has infinitely many points.

So a circle has **infinitely many tangents**.

2 (i) **Fill in the blank: A tangent to a circle intersects it in _____ point(s).**

✔ ANSWER

one point.

2 (ii) **Fill in the blank: A line intersecting a circle in two points is called a _____.**

✔ ANSWER

secant.

2 (iii) **Fill in the blank: A circle can have _____ parallel tangents at the most.**

✔ ANSWER

two parallel tangents (at the two ends of a diameter).

2 (iv) **Fill in the blank: The common point of a tangent to a circle and the circle is called**

_____.

✔ ANSWER

point of contact.

3 **A tangent PQ at a point P of a circle of radius 5 cm meets a line through the centre O at a point Q so that OQ = 12 cm. Length PQ is: (A) 12 cm (B) 13 cm (C) 8.5 cm (D) √119 cm**

✔ ANSWER

Option (D) √119 cm

OP is the radius through the point of contact P, so OP ⊥ PQ (Theorem 10.1).

In right ΔOPQ, by Pythagoras theorem:

$PQ^2 = OQ^2 - OP^2 = 12^2 - 5^2 = 144 - 25 = 119$

PQ = √119 cm

4 **Draw a circle and two lines parallel to a given line such that one is a tangent and the other, a secant to the circle.**

✔ ANSWER

Steps:

- Draw a circle with centre O and any line l outside it (or anywhere).
- Draw a line m parallel to l that cuts the circle at two points A and B. This line is a **secant**.

- Draw a line n parallel to l that touches the circle at only one point P (for this, draw the perpendicular from O to l; let it meet the circle at P; draw the line through P parallel to l). This line is a **tangent**.

Result: lines m and n are both parallel to l; m is a secant and n is a tangent to the circle.

Exercise 10.2

1 **From a point Q, the length of the tangent to a circle is 24 cm and the distance of Q from the centre is 25 cm. The radius of the circle is (A) 7 cm (B) 12 cm (C) 15 cm (D) 24.5 cm**

✔ ANSWER

Option (A) 7 cm

Let the tangent from Q touch the circle at P. Then OP ⊥ PQ (Theorem 10.1).

In right ΔOPQ: $OP^2 = OQ^2 - PQ^2 = 25^2 - 24^2 = 625 - 576 = 49$

OP = 7 cm

2 **In Fig. 10.11, if TP and TQ are the two tangents to a circle with centre O so that ∠POQ = 110°, then ∠PTQ is equal to (A) 60° (B) 70° (C) 80° (D) 90°**

✔ ANSWER

Option (B) 70°

OP ⊥ TP and OQ ⊥ TQ, so ∠OPT = ∠OQT = 90°.

In quadrilateral OPTQ, the sum of angles is 360°:

∠PTQ = 360° − (90° + 90° + 110°) = 360° − 290°

∠PTQ = 70°

3 **If tangents PA and PB from a point P to a circle with centre O are inclined to each other at angle of 80°, then ∠POA is equal to (A) 50° (B) 60° (C) 70° (D) 80°**

✔ ANSWER

Option (A) 50°

PA = PB (tangents from an external point), OA = OB (radii), OP = OP (common).

So ΔOAP ≅ ΔOBP (SSS), giving ∠APO = ∠BPO = ½ × 80° = 40°.

In ΔOAP, ∠OAP = 90° (radius ⊥ tangent).

∠POA = 180° − 90° − 40° = **50°**

4 **Prove that the tangents drawn at the ends of a diameter of a circle are parallel.**

✔ ANSWER

Given: a circle with centre O, diameter AB, tangent PQ at A and tangent RS at B.

To prove: PQ ∥ RS.

Proof:

1. OA ⊥ PQ (radius through the point of contact is perpendicular to the tangent, Theorem 10.1). So ∠OAQ = 90°.

2. OB ⊥ RS. So ∠OBR = 90°.

3. ∠OAQ and ∠OBR are alternate interior angles for the lines PQ and RS with the transversal AB, and they are equal (90° each).

4. Hence **PQ ∥ RS**. (Also: ∠OAQ + ∠OBS = 180°, co-interior angles.)

5 **Prove that the perpendicular at the point of contact to the tangent to a circle passes through the centre.**

✔ ANSWER

Given: a circle with centre O, tangent AB at point P, and a line l perpendicular to AB at P.

To prove: l passes through O.

Proof (by contradiction):

1. Suppose l does not pass through O. Join OP.

2. OP ⊥ AB (radius through the point of contact is perpendicular to the tangent, Theorem 10.1). So ∠OPB = 90°.

3. Also l ⊥ AB at P, so the angle between l and AB at P is 90°.

4. Thus OP and l are two different lines through P, both perpendicular to AB. But only one perpendicular can be drawn to a line at a given point on it.

5. So OP and l are the same line, i.e. l passes through O.

Hence the perpendicular at the point of contact to the tangent passes through the centre.

6 **The length of a tangent from a point A at distance 5 cm from the centre of the circle is 4 cm. Find the radius of the circle.**

✔ ANSWER

Let the tangent from A touch the circle at P, centre O.

Given: OA = 5 cm, AP = 4 cm. OP ⊥ AP (Theorem 10.1).

In right ΔOPA: $OP^2 = OA^2 - AP^2 = 5^2 - 4^2 = 25 - 16 = 9$

Radius OP = 3 cm

7 **Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.**

✔ ANSWER

Let O be the common centre and AB the chord of the larger circle touching the smaller circle at P. Join OP, OA.

OP = 3 cm, OA = 5 cm. OP ⊥ AB (radius ⊥ tangent), so OP bisects the chord AB.

In right ΔOPA: $AP^2 = OA^2 - OP^2 = 25 - 9 = 16$, so AP = 4 cm.

AB = 2 × AP = **8 cm**

8 **A quadrilateral ABCD is drawn to circumscribe a circle (see Fig. 10.12). Prove that AB + CD = AD + BC.**

✔ ANSWER

Let the circle touch AB, BC, CD, DA at P, Q, R, S respectively.

Tangents drawn from an external point to a circle are equal (Theorem 10.2):

- AP = AS (from A)
- BP = BQ (from B)
- CR = CQ (from C)
- DR = DS (from D)

Adding: AP + BP + CR + DR = AS + BQ + CQ + DS

(AP + BP) + (CR + DR) = (AS + DS) + (BQ + CQ)

AB + CD = AD + BC

9 **In Fig. 10.13, XY and X′Y′ are two parallel tangents to a circle with centre O and another tangent AB with point of contact C intersecting XY at A and X′Y′ at B. Prove that ∠AOB = 90°.**

✔ ANSWER

Let XY touch the circle at P and X′Y′ at Q. Join OP, OQ, OC.

1. In ΔOPA and ΔOCA: OP = OC (radii), AP = AC (tangents from A), OA = OA. So ΔOPA ≅ ΔOCA (SSS), hence ∠POA = ∠COA … (1)

2. Similarly ΔOQB ≅ ΔOCB, hence ∠QOB = ∠COB … (2)

3. POQ is a straight line (OP ⊥ XY, OQ ⊥ X′Y′ and XY ∥ X′Y′, so PQ is a diameter). So ∠POA + ∠COA + ∠COB + ∠QOB = 180°.

4. Using (1) and (2): 2∠COA + 2∠COB = 180°, so ∠COA + ∠COB = 90°.

∠AOB = 90°

10 **Prove that the angle between the two tangents drawn from an external point to a circle is supplementary to the angle subtended by the line-segment joining the points of contact at the centre.**

✔ ANSWER

Given: circle with centre O; tangents PA and PB from an external point P; A, B points of contact.

To prove: ∠APB + ∠AOB = 180°.

Proof:

1. OA ⊥ PA and OB ⊥ PB (Theorem 10.1). So ∠OAP = ∠OBP = 90°.

2. In quadrilateral OAPB, ∠OAP + ∠APB + ∠PBO + ∠AOB = 360°.

3. 90° + ∠APB + 90° + ∠AOB = 360°

4. **∠APB + ∠AOB = 180°**, i.e. the two angles are supplementary.

11 **Prove that the parallelogram circumscribing a circle is a rhombus.**

✔ ANSWER

Let ABCD be a parallelogram circumscribing a circle, touching AB, BC, CD, DA at P, Q, R, S.

1. Tangents from an external point are equal: AP = AS, BP = BQ, CR = CQ, DR = DS.

2. Adding: AB + CD = AD + BC (as in Q8).

3. In a parallelogram, AB = CD and AD = BC. So 2AB = 2BC, i.e. AB = BC.

4. Hence AB = BC = CD = DA.

A parallelogram with all sides equal is a rhombus.

12 **A triangle ABC is drawn to circumscribe a circle of radius 4 cm such that the segments BD and DC into which BC is divided by the point of contact D are of lengths 8 cm and 6 cm respectively (see Fig. 10.14). Find the sides AB and AC.**

✔ ANSWER

Let the circle (centre O, radius 4 cm) touch AB at F and AC at E. Let AF = AE = x (tangents from A).

Tangents from B: BF = BD = 8 cm. Tangents from C: CE = CD = 6 cm.

So AB = x + 8, AC = x + 6, BC = 8 + 6 = 14 cm.

Area of ΔABC = Area ΔOBC + Area ΔOCA + Area ΔOAB (OD, OE, OF ⊥ sides, each = 4 cm)

= ½ × 4 × (14 + x + 6 + x + 8) = 2(28 + 2x) = 56 + 4x ... (1)

By Heron's formula: s = (14 + x + 8 + x + 6)/2 = 14 + x

Area = √[s(s − a)(s − b)(s − c)] = √[(14 + x)(14 + x − 14)(14 + x − x − 6)(14 + x − x − 8)]

= √[(14 + x) · x · 8 · 6] = √[48x(14 + x)] ... (2)

From (1) and (2): 56 + 4x = √[48x(14 + x)]

Squaring: $(56 + 4x)^2 = 48x(14 + x)$

$16(14 + x)^2 = 48x(14 + x)$

Dividing by 16(14 + x): 14 + x = 3x, so 2x = 14, **x = 7**

AB = x + 8 = 15 cm, AC = x + 6 = 13 cm

Check: sides 15, 13, 14; area = 84 cm² both ways; inradius = 84/42 = 4 cm ✓

13 **Prove that opposite sides of a quadrilateral circumscribing a circle subtend supplementary angles at the centre of the circle.**

✔ ANSWER

Let ABCD circumscribe a circle with centre O, touching AB, BC, CD, DA at P, Q, R, S. Join OP, OQ, OR, OS, OA, OB, OC, OD.

To prove: ∠AOB + ∠COD = 180° and ∠BOC + ∠AOD = 180°.

1. The two tangents from an external point are equally inclined to the line joining the point to the centre (Theorem 10.2 remark), so ΔOAP ≅ ΔOAS, ΔOBP ≅ ΔOBQ, ΔOCQ ≅ ΔOCR, ΔODR ≅ ΔODS.

2. Hence ∠1 = ∠8, ∠2 = ∠3, ∠4 = ∠5, ∠6 = ∠7, where ∠1 = ∠AOP, ∠2 = ∠BOP, ∠3 = ∠BOQ, ∠4 = ∠COQ, ∠5 = ∠COR, ∠6 = ∠DOR, ∠7 = ∠DOS, ∠8 = ∠AOS.

3. All eight angles around O add to 360°: ∠1 + ∠2 + … + ∠8 = 360°.

4. So 2(∠1 + ∠2 + ∠5 + ∠6) = 360°, i.e. (∠1 + ∠2) + (∠5 + ∠6) = 180°.

5. ∠1 + ∠2 = ∠AOB and ∠5 + ∠6 = ∠COD. So **∠AOB + ∠COD = 180°**.

6. Similarly 2(∠3 + ∠4 + ∠7 + ∠8) = 360° gives **∠BOC + ∠AOD = 180°**.

Hence opposite sides subtend supplementary angles at the centre.

🎉 **Chapter 10 done! Revise the answers once more tomorrow.**